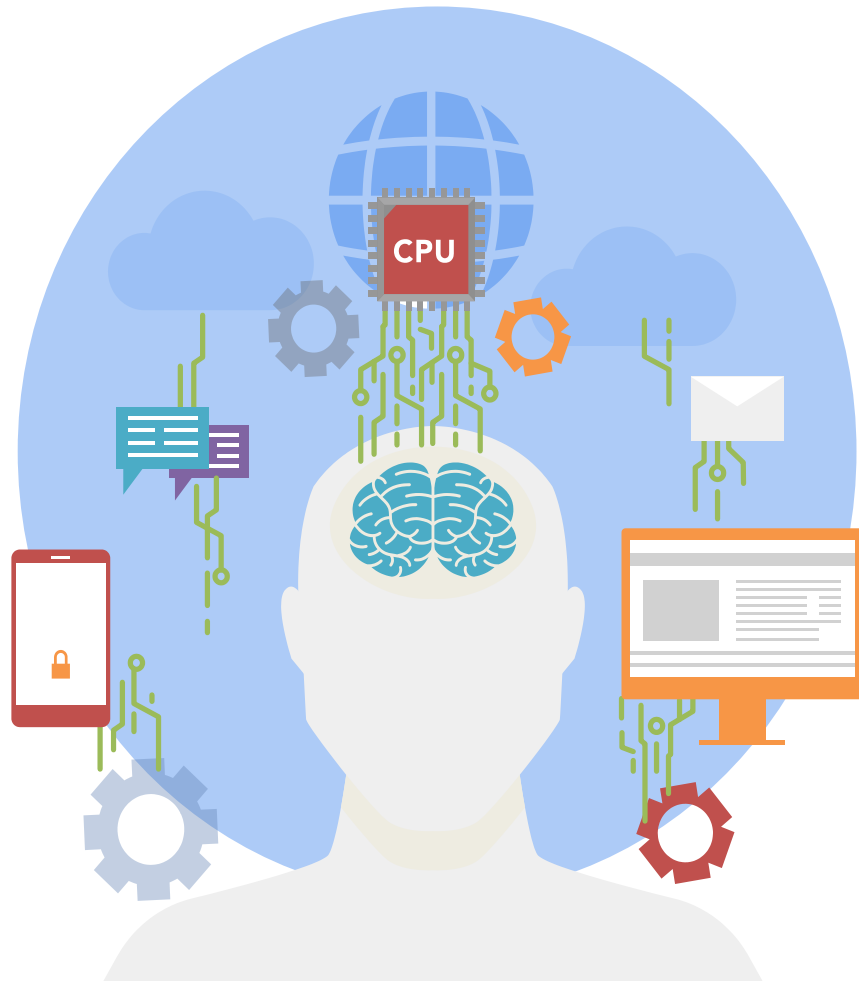




# Understanding Agentic AI

Dr. Tamer Arafa



# Introduction to Agentic Workflows

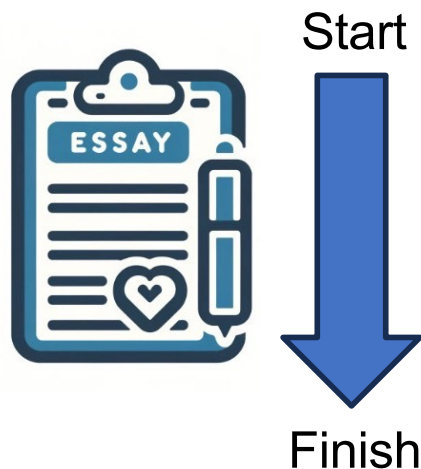
---

## What is Agentic AI?

# Agentic AI

Non-agentic workflow (zero-shot):

Please type out an essay on topic X from start to finish in one go, without using backspace.



Agentic workflow:

Write an essay outline on topic X

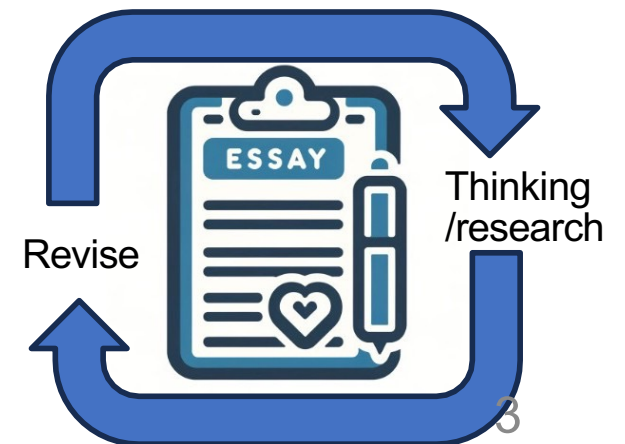
Do you need any web research?

Write a first draft.

Consider what parts need revision or more research.

Revise your draft.

...



# Agentic AI workflows

An agentic AI workflow is a process where an LLM-based app executes multiple steps to complete a task.

## Essay-writing example:

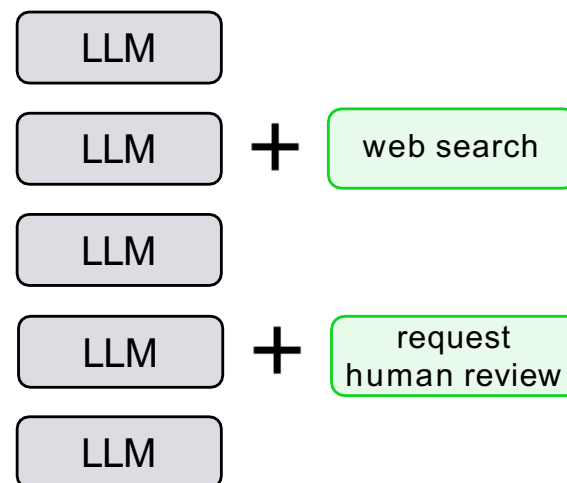
Write an essay outline on topic X

Do you need any web research?

Write a first draft.

Consider what parts need revision or more research.

Revise your draft.





Sequoia Ascent, March 2024



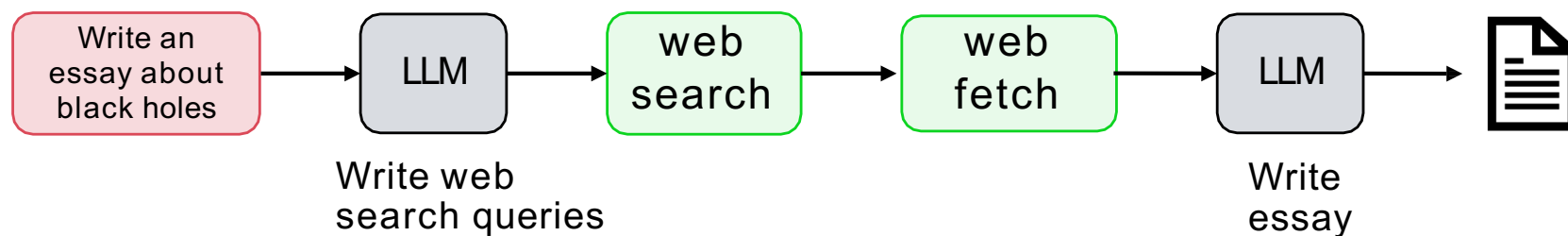
....  
Rather than arguing over which work to include or exclude as being a true agent, we can acknowledge that there are different degrees to which systems can be agentic.  
....



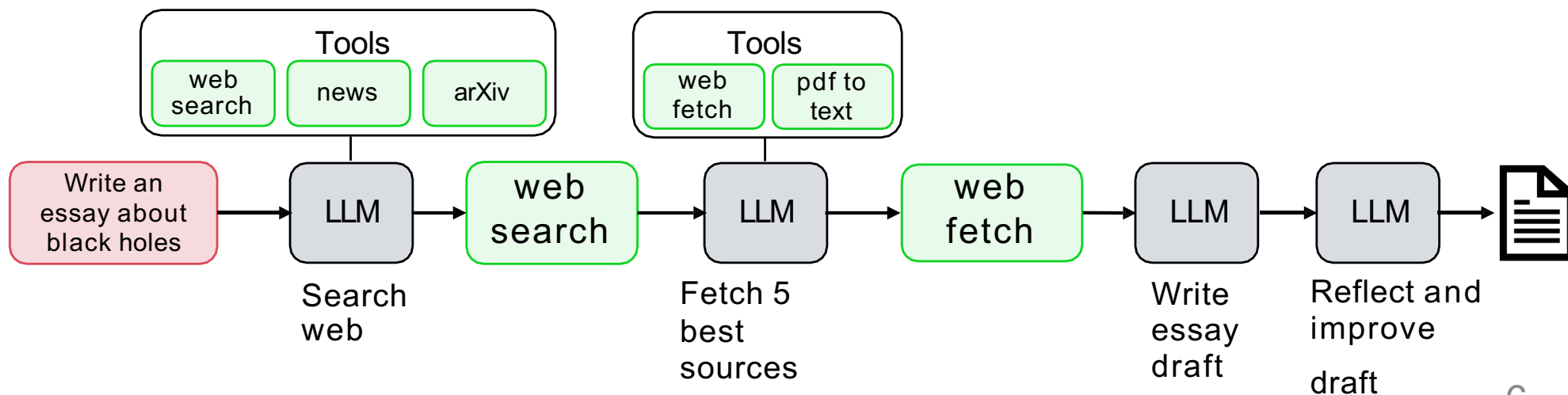
X (twitter) post , June 2024

# Degrees of autonomy

Less autonomous

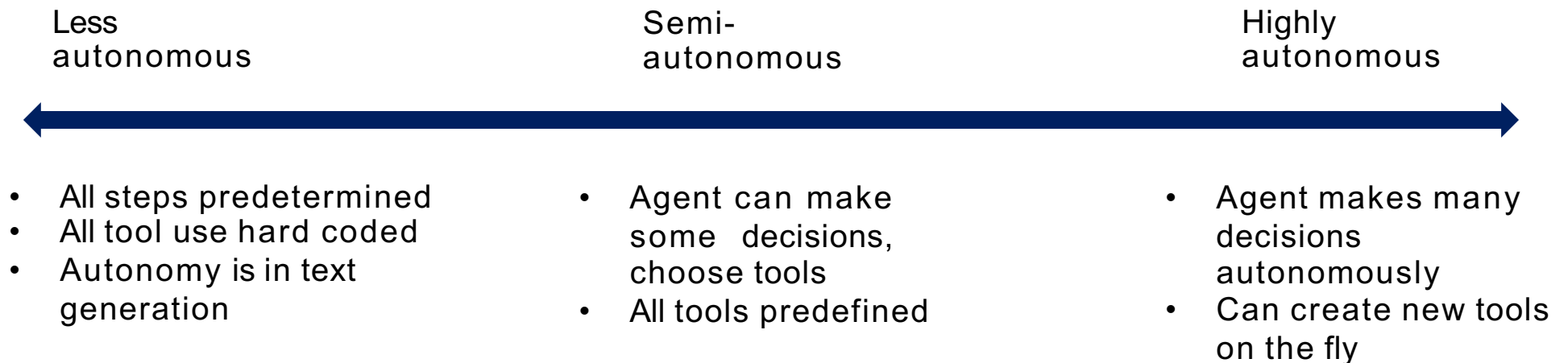


More autonomous

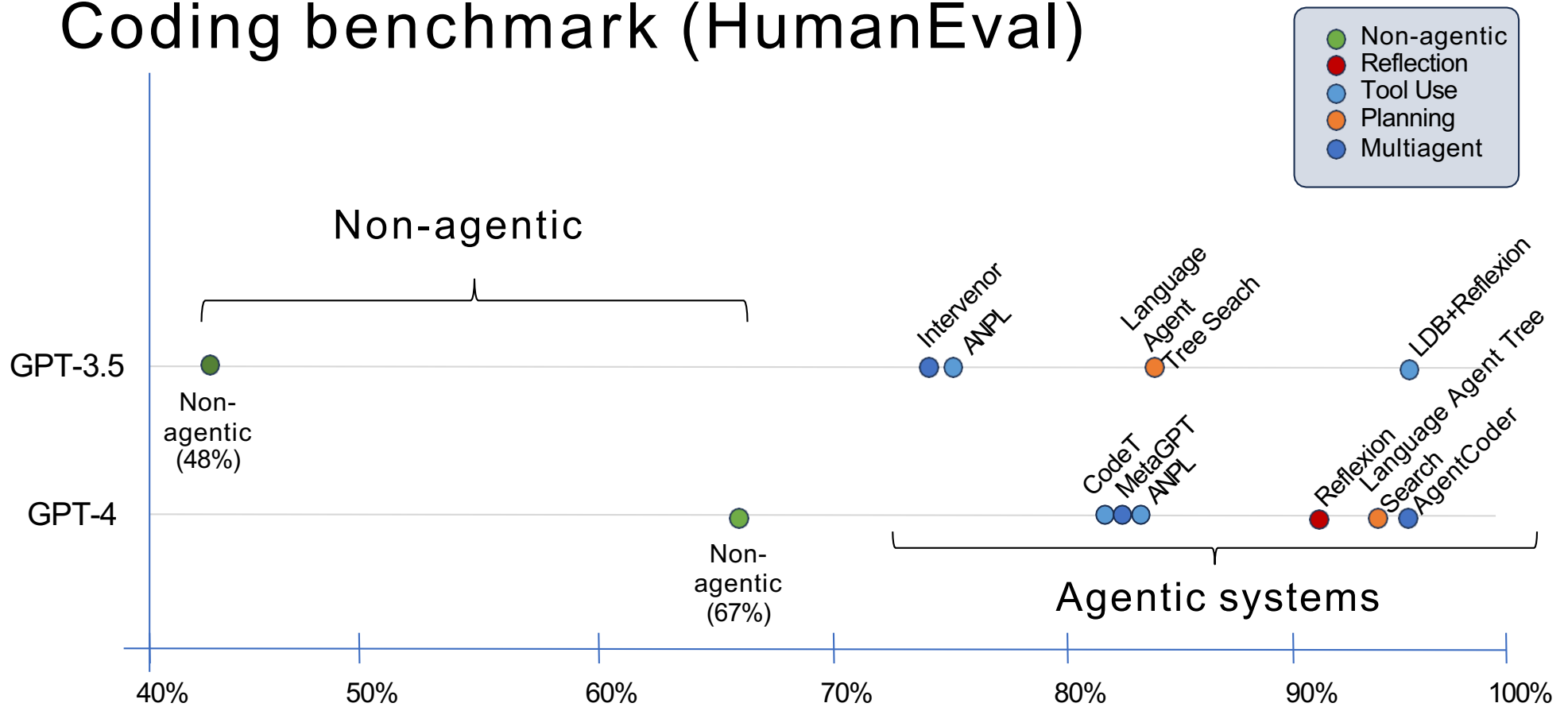


# Degrees of “agenticness”

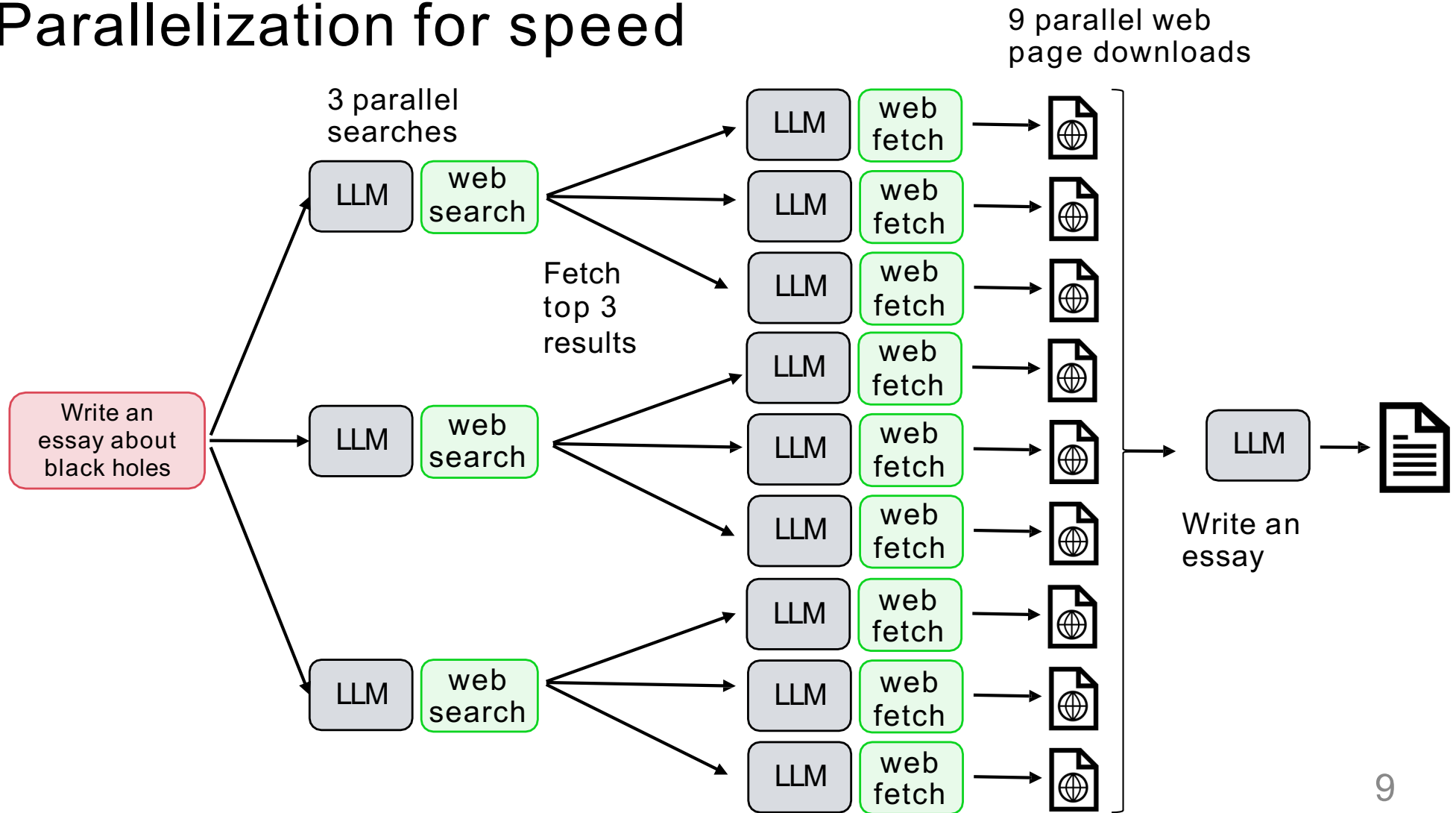
Agentic AI can be less or more autonomous



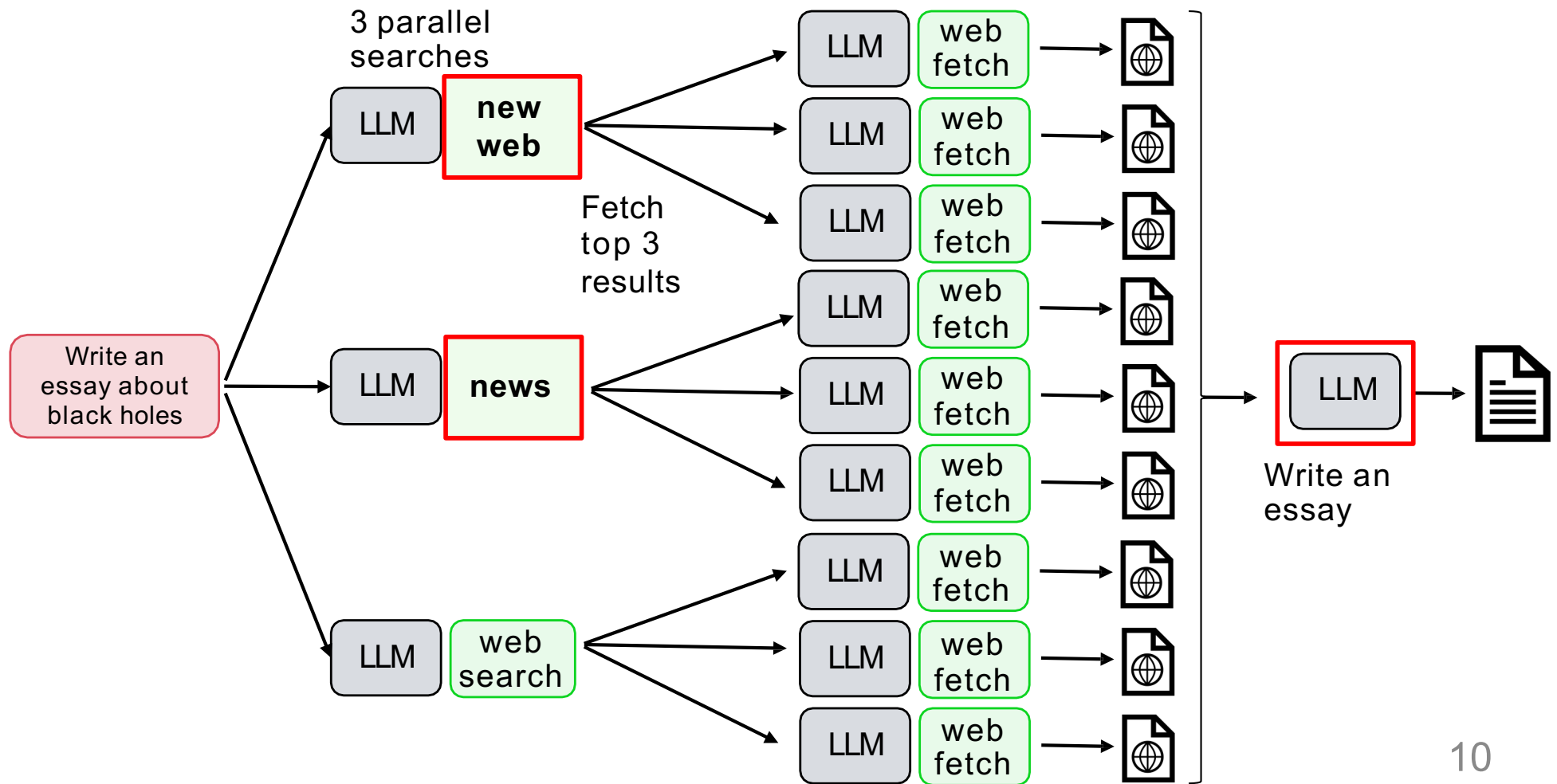
# Coding benchmark (HumanEval)



# Parallelization for speed

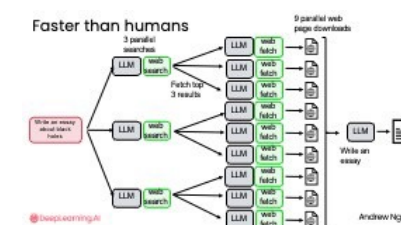
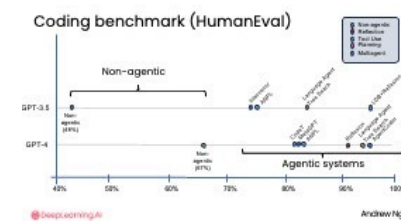


## Modular: Add/swap out components



# Key benefits of agentic workflows

- Much better performance
- Faster than humans because of parallelization
- Modular: can add or update tools, swap out models





# Introduction to Agentic Workflows

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Agentic AI applications

# Example: Invoice processing workflow

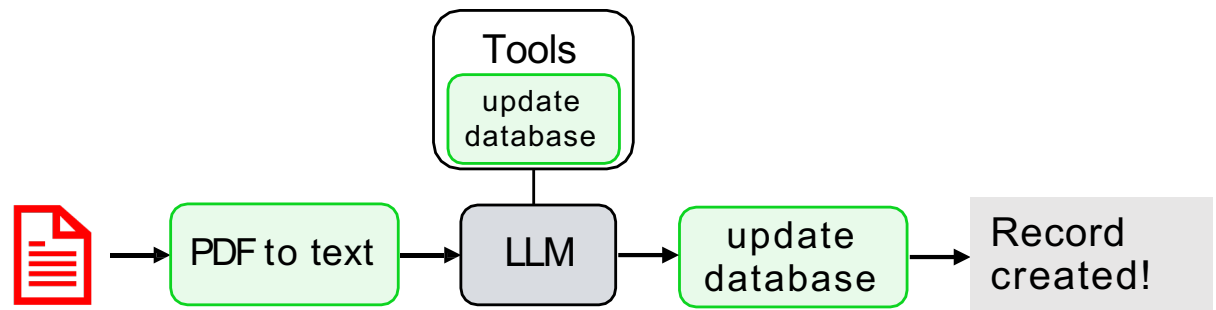
| <b>TechFlow Solutions LLC</b><br>890 Juniper Drive<br>San Mateo, CA 94401<br>Phone: (415) 555-7890<br>Email: billing@techflowsol.com |     |                                     |                   |
|--------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------|-------------------|
| <b>Due Date: August 20, 2025</b>                                                                                                     |     | <b>Invoice Date: August 6, 2025</b> |                   |
| Description                                                                                                                          | Qty | Unit Price                          | Line Total        |
| Consulting - Systems Integration (hrs)                                                                                               | 20  | \$150.00                            | \$3,000.00        |
| <b>Total Due:</b>                                                                                                                    |     |                                     | <b>\$3,000.00</b> |

## 4 required fields:

Biller  
Biller address  
Amount due  
Due date

## Steps:

1. Identify required fields
2. Record in database



# Example: Responding to customer email

From: Susan Jones  
Subject: Wrong item shipped

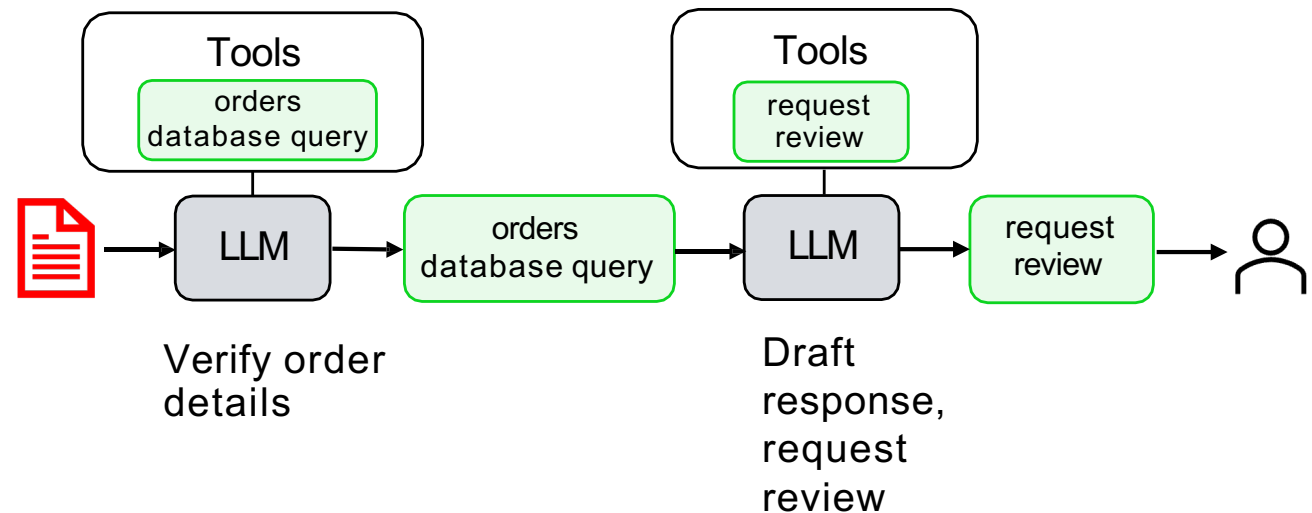
I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan

Steps:

1. Extract key information
2. Find relevant customer records
3. Draft response for human review



# More challenging: Customer service agent

Do you have any black jeans or blue jeans?

1. Check inventory for black jeans
2. Check inventory for blue jeans
3. Respond to customer

I'd like to return the beach towel I bought

1. Verify customer purchase
2. Check return policies
3. If return allowed = "yes", then:
  - a. Issue return packing slip
  - b. Set database record to "return pending"

Required steps  
not known ahead  
of time

# What tasks is agentic AI suited to?

Easier

Harder



Clear, step-by-step process

Steps not known ahead of time

Standard procedures to follow

Plan/solve as you go

Text assets only

Multimodal (sound, vision)



# Introduction to Agentic Workflows

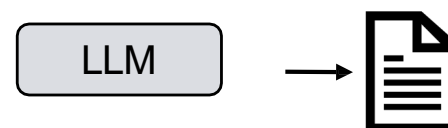
---

Task decomposition:  
Identifying the steps in  
a workflow

# Example: Writing an Essay

Write an essay on topic X

1. Write an essay outline on topic X
2. Search web
3. ~~Write the essay~~ Write a first draft
4. Consider what parts need revision
5. Revise your draft



# Recap: Breaking down essay writing into steps

Direct generation:

1. Write an essay on topic X

Not good enough...

3-step workflow:

1. Write an essay outline on topic X
2. Search web
3. Write the essay

Still not good enough...

5-step workflow:

1. Write an essay outline on topic X
2. Search web
3. Write the first draft
4. Consider what parts need revision
5. Revise your draft

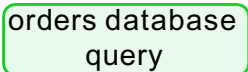
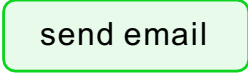
# Example: Responding to customer email

From: [sjones9@email.com](mailto:sjones9@email.com)  
Subject: Wrong item shipped

Hi, I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan Jones

1. Extract key information 
2. Find relevant customer records  + 
3. Write and send response  + 

# Example: Extracting information from invoice

| <b>TechFlow Solutions LLC</b><br>890 Juniper Drive<br>San Mateo, CA 94401<br>Phone: (415) 555-7890<br>Email: billing@techflowsol.com |            |                                     |                   |
|--------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------|-------------------|
| <b>Due Date: August 20, 2025</b>                                                                                                     |            | <b>Invoice Date: August 6, 2025</b> |                   |
| <b>Description</b>                                                                                                                   | <b>Qty</b> | <b>Unit Price</b>                   | <b>Line Total</b> |
| Consulting - Systems Integration (hrs)                                                                                               | 20         | \$150.00                            | \$3,000.00        |
| <b>Total Due:</b>                                                                                                                    |            |                                     | <b>\$3,000.00</b> |

1. Find required information

LLM

LLM

+

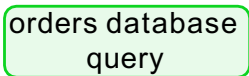
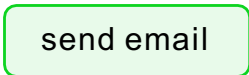
update  
database

# What building blocks do you have?

| Building block | Examples              | Use cases                                                       |
|----------------|-----------------------|-----------------------------------------------------------------|
| Models         | LLMs                  | Text generation, tool use, information extraction               |
|                | Other AI models       | PDF-to-text, text-to-speech, image analysis                     |
| Tools          | API                   | Web search, get real-time data, send email, check calendar,.... |
|                | Information retrieval | Databases, Retrieval Augmented Generation (RAG)                 |
|                | Code execution        | Basic calculator, data analysis                                 |

# Tips for task decomposition

Example: Responding to email

1. Extract key information 
2. Find relevant customer records  + 
3. Write and send response  + 

Questions to ask:

- Can LLM, tool, or both complete each step?
- If “no”, consider breaking down into further steps, or adding human-in-the-loop



# Introduction to Agentic Workflows

---

## Evaluating Agentic AI (evals)

# Look for low-quality outputs

From: [sjones9@email.com](mailto:sjones9@email.com)  
Subject: Wrong item shipped

Hi, I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan Jones

1. Extract key information

LLM

2. Find relevant customer records

LLM

+

orders database query

3. Write and send response

LLM

+

send email

“I’m glad you shopped with us – we’re much better than our competitor CompCo!”

“Sure, we’ll issue a refund. Unlike RivalCo, we make returns easy.”

# Add an evaluation to track the error

Customer  
inquiry

We're better  
than CompCo...

LLM response

```
if (competitor in response):  
    num_competitor_mentions += 1
```

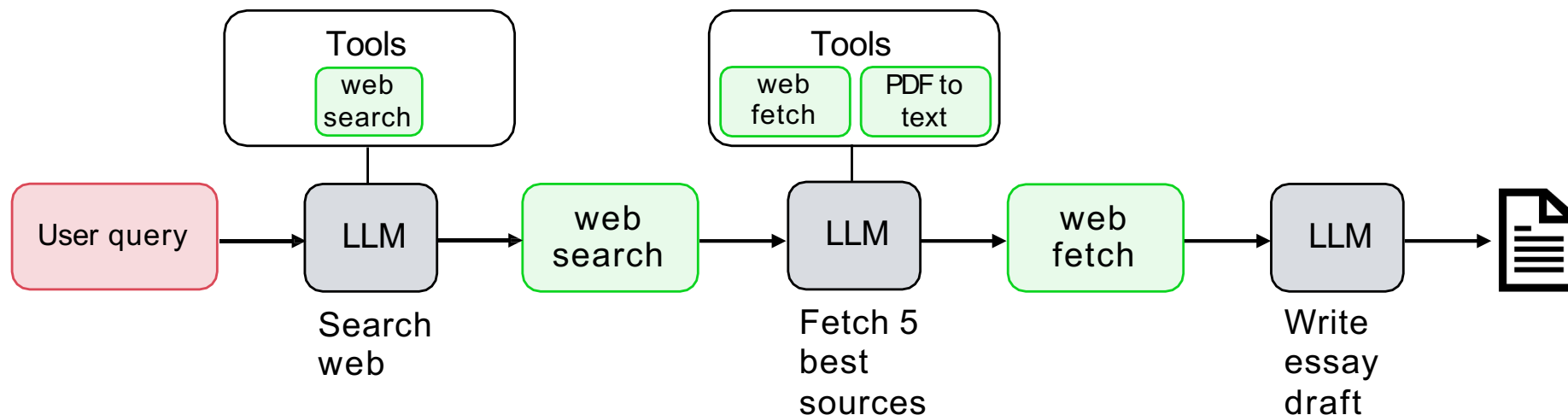
List of competitors:

CompCo  
RivalCo  
The Other Co  
...

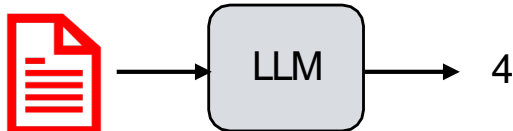
Objective error:

Use code to check for occurrences

# Using LLM as a judge



Assign the following essay a quality score between 1 and 5, where 5 is the best: {essay}



| Example prompt     | LLM-assigned score |
|--------------------|--------------------|
| Black holes        | 3                  |
| Robotic harvesting | 4                  |

# Evaluating Agentic AI

- Can evaluate using code (objective evals), or LLM-as-judge (subjective evals)
- Two types of evals: End-to-end and component-level
- Examine traces to perform error analysis
- Much more on evals and error analysis in Module 4!



# Introduction to Agentic Workflows

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Agentic design patterns

# Agentic Design Patterns

1. Reflection
2. Tool use
3. Planning
4. Multi-agent collaboration

# Reflection



Please write code for {task}

`def do_task(x): ...`

`def do_task_v2(x):`

`def do_task_v3(x):`



Coder  
Agent  
(LLM)

Here's code intended for {task}:

```
def do_task (x):
```

```
...
```

Check the code carefully for correctness, style and efficiency, and give constructive criticism for how to improve it.

There's a bug on line 5. Fix it by ...

It failed Unit Test 3. Try changing ...

# Reflection



Please write code for {task}

`def do_task(x): ...`

`def do_task_v2(x):`

`def do_task_v3(x):`



Coder  
Agent  
(LLM)

Here's code intended for {task}:

```
def do_task (x):
```

```
...
```

Check the code carefully for correctness, style and efficiency, and give constructive criticism for how to improve it.

There's a bug on line 5. Fix it by ...

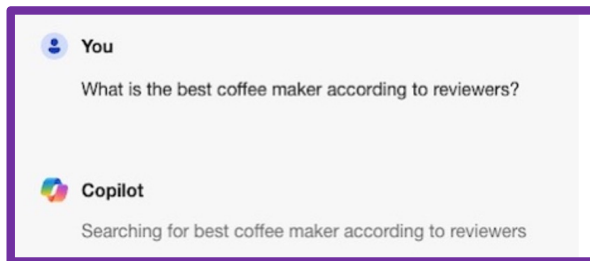
It failed Unit Test 3. Try changing ...



Critic  
Agent  
(LLM)

# Tool use

## Web search tool



## Code execution tool



### Analysis

- Code Execution
- Wolfram Alpha
- Bearly Code Interpreter

### Information gathering

- Web search
- Wikipedia
- Database access

### Productivity

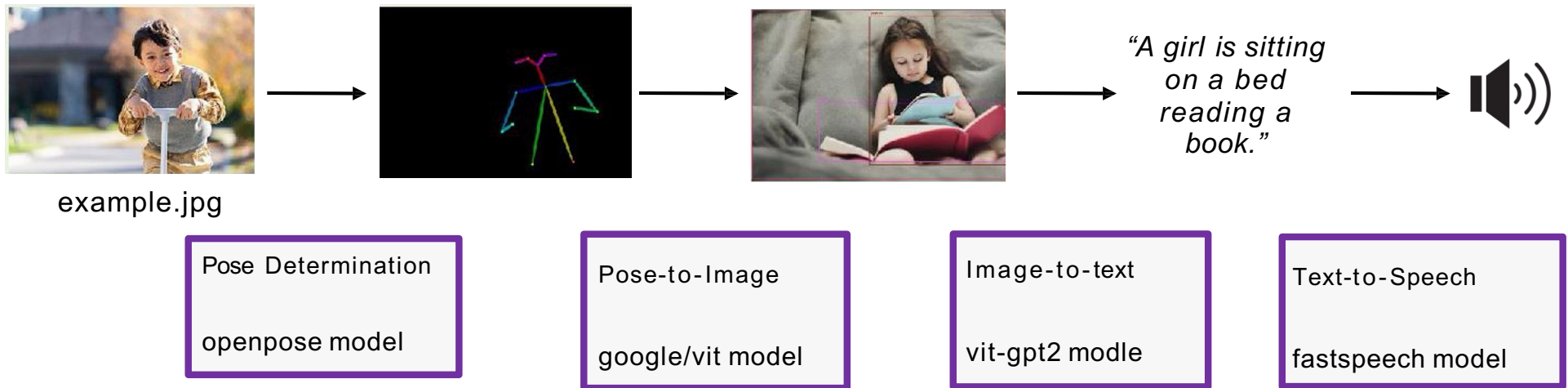
- Email
- Calendar
- Messaging

### Images

- Image generation
- Image captioning
- OCR

# Planning

Request: Please generate an image where a girl is reading a book, and her pose is the same as the boy in the image example.jpg, then please describe the new image with your voice.



[Example adapted from "HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face", Shen et al. (2023)]

# Multi-agentic workflows



## Multiagent Debate

| Task        | Single agent | Multi-agent  |
|-------------|--------------|--------------|
| Biographies | 66.0%        | <b>73.8%</b> |
| MMLU        | 63.9%        | <b>71.1%</b> |
| Chess move  | 29.3%        | <b>45.2%</b> |

[From “Improving Factuality and Reasoning in Language Models through Multiagent Debate”, Du et al., 2023]

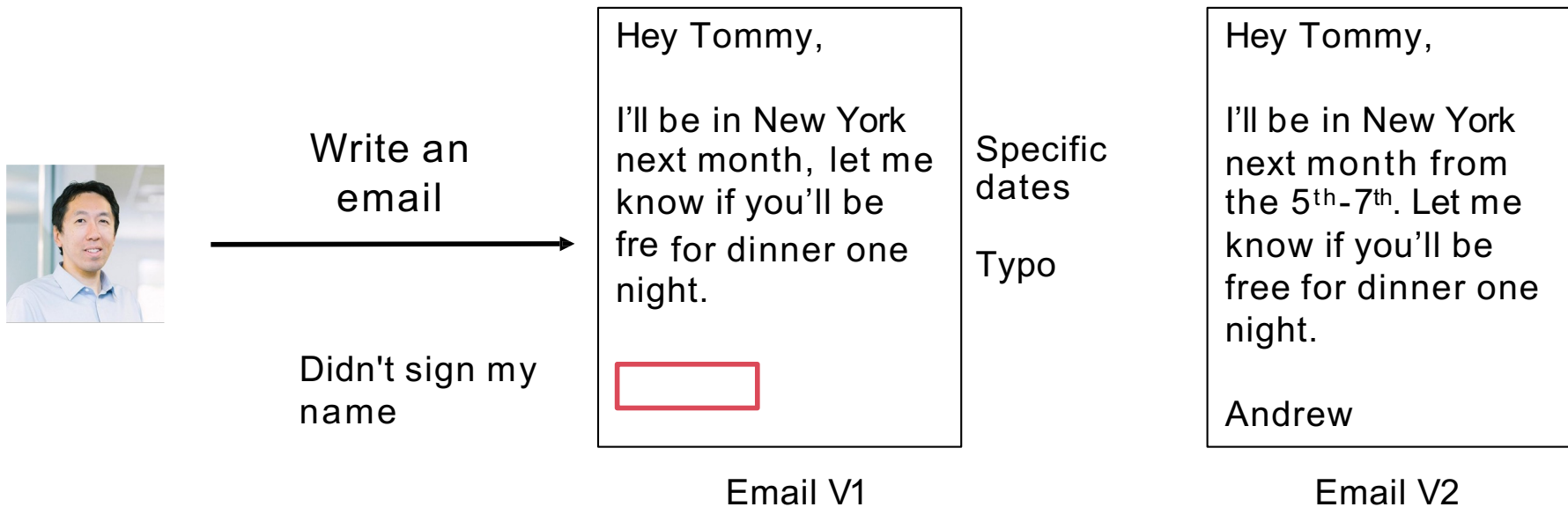


# Reflection Design Pattern

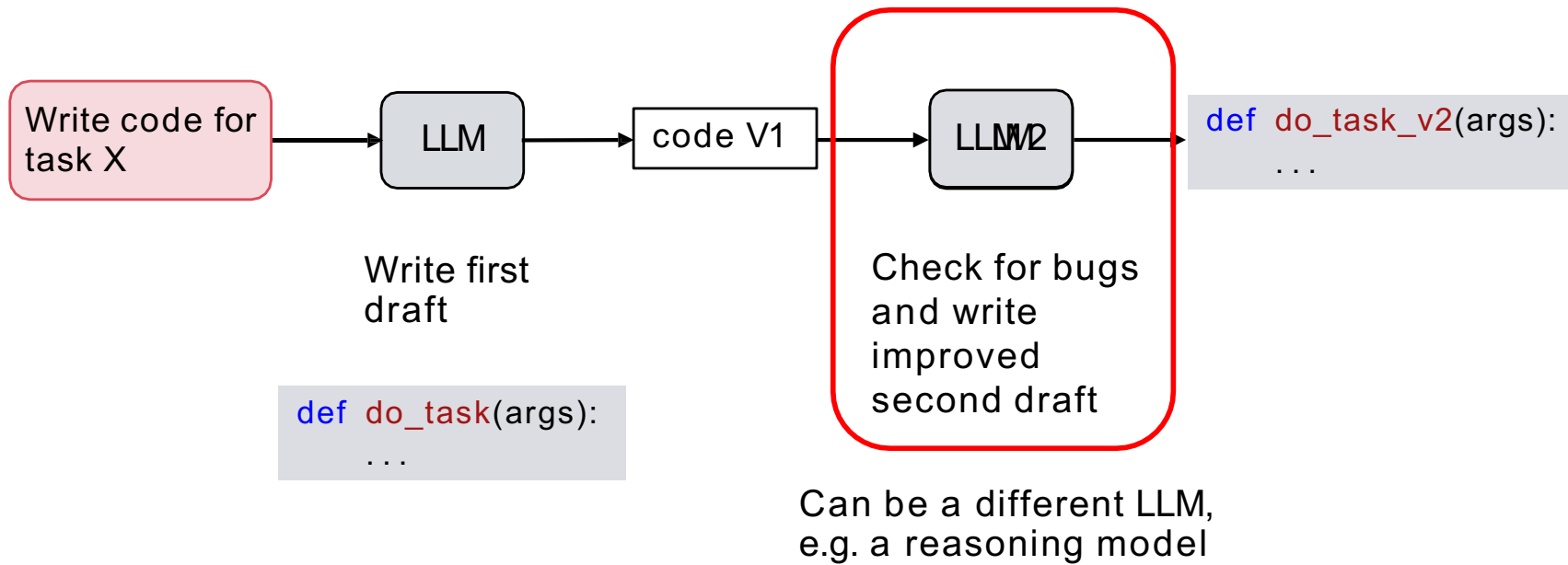
---

Reflection to improve  
outputs of a task

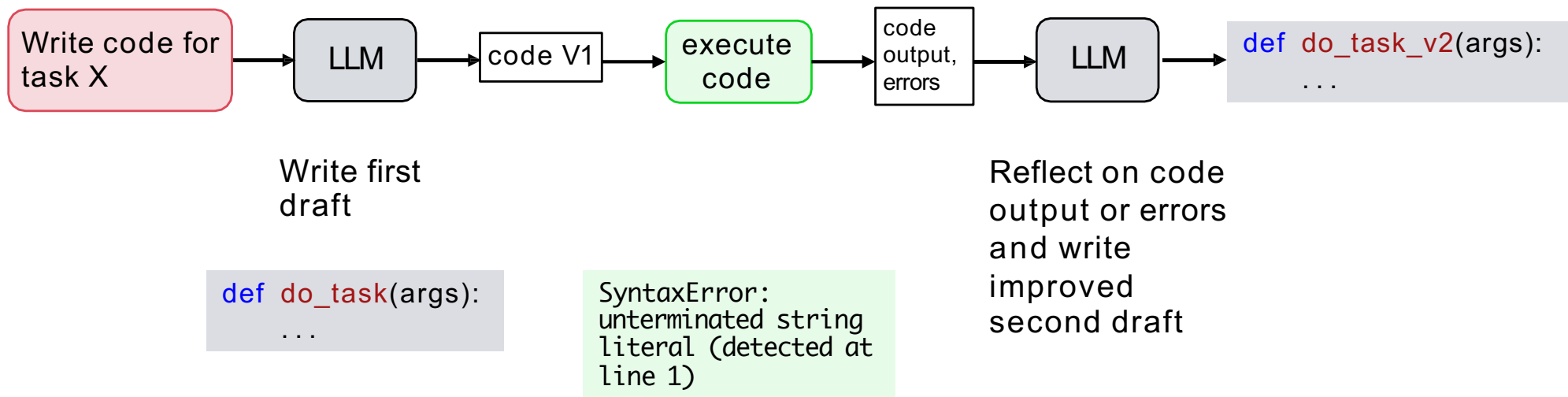
# Reflection - humans

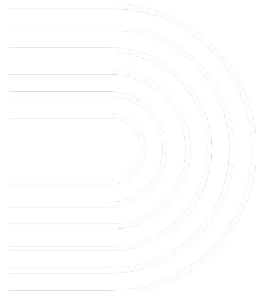


# Reflection to improve code



# Reflection with external feedback



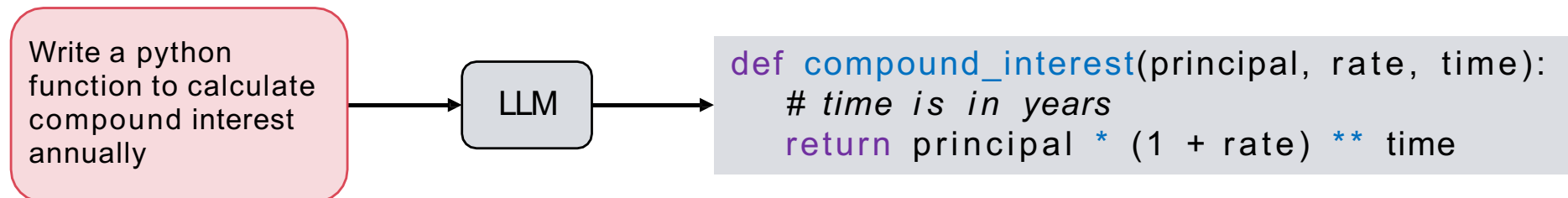


# Reflection Design Pattern

---

Why not just direct  
generation?

# Direct generation



# Zero, one, and few-shot prompting

Convert to  
MM/DD/YYYY format

Input:  
{input\_date}

Zero-shot (no examples)

Convert to  
MM/DD/YYYY format

Input: Jan 1<sup>st</sup>, 2025  
Output: 01/01/2025

Input:  
{input\_date}

One-shot (single example)

Convert to  
MM/DD/YYYY format

Input: Jan 1<sup>st</sup>, 2025  
Output: 01/01/2025

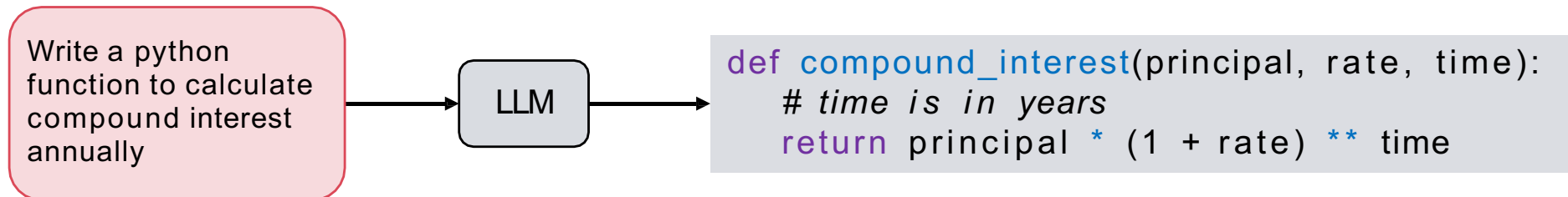
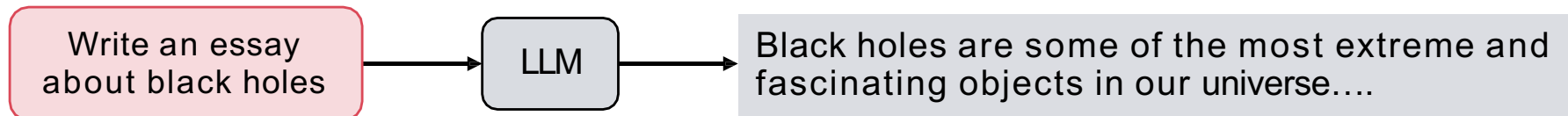
Input: 21<sup>st</sup> June, 2025  
Output: 06/21/2025

Input:  
{input\_date}

Two-shot (two examples)  
Few-shot (multiple examples)

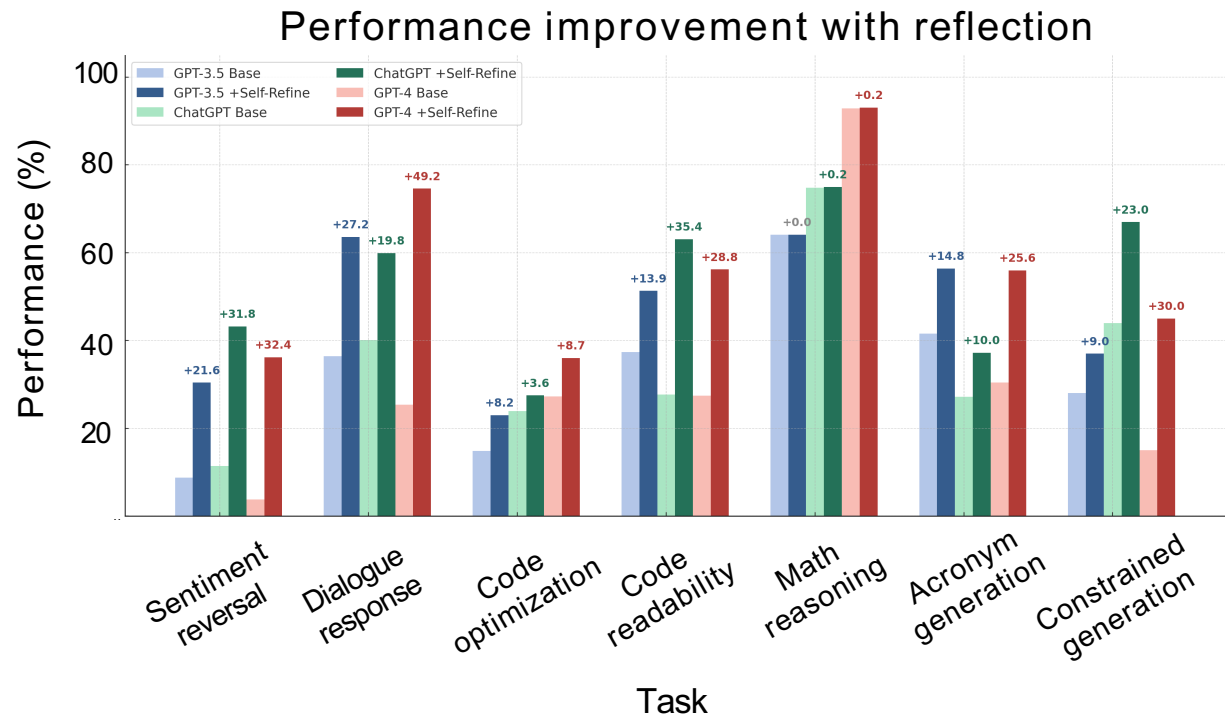
# Direct generation

## Zero-shot prompts



# Reflection has been tested

Reflection consistently outperforms direct generation on a variety of tasks.



[Adapted from Madaan, A. et al. (2023) "Self-refine: Iterative refinement with self-feedback"]

# Tasks where reflection works better

| Example                          | Problem                                              | Reflection prompt                                                                      |
|----------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------|
| Generate html table              | Missing '>'                                          | Validate the html code                                                                 |
| How to brew a perfect cup of tea | Missing steps                                        | Check instructions for coherence and completeness                                      |
| Generating domain names          | Name has unintended meaning, or is hard to pronounce | Does domain name have any negative connotations? Is the domain name hard to pronounce? |

# Tips for writing reflection prompts

## Brainstorming domain names

Review the domain names you suggested.

Check if each name is easy to pronounce and thus easy to spread via word of mouth.

Consider whether each name might mean something negative in other languages.

Then output a shortlist of only the names that meet these criteria.

## Improving email

Review the email first draft.

Check that the tone is professional and look for phrases that could be considered rude or insensitive.

Verify all facts, dates, and promises are accurate.

Then write the next draft of the email.

- Clearly indicate the reflection action
- Specify criteria to check

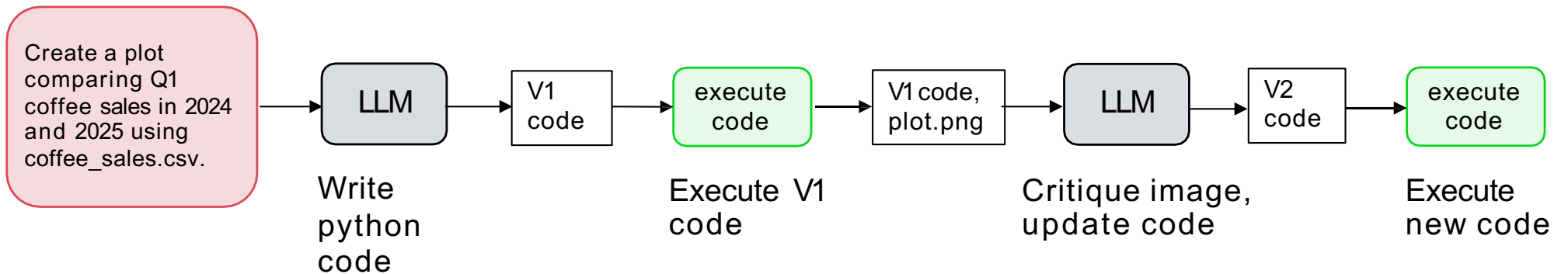
# Visualizing coffee sales



| date       | price | coffee_name   |
|------------|-------|---------------|
| 2024-01-12 | 3.87  | Latte         |
| 2024-01-28 | 3.87  | Hot Chocolate |
| 2024-02-09 | 3.87  | Hot Chocolate |
| 2024-03-01 | 2.89  | Cappuccino    |
| 2024-03-04 | 3.87  | Latte         |
| ...        | ...   | ...           |
| 2025-03-23 | 3.57  | Latte         |

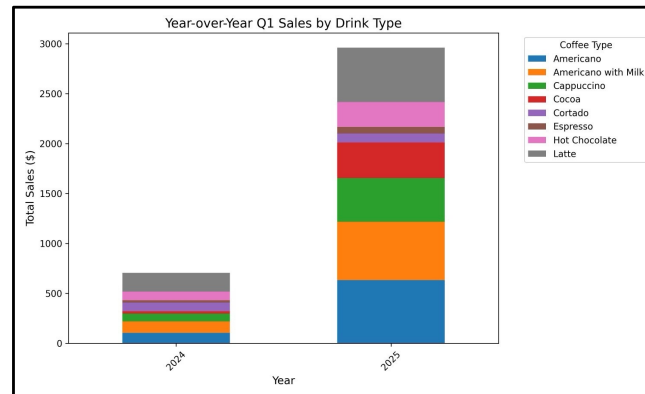
Create a plot comparing Q1 coffee sales in 2024 and 2025

# Chart generation agentic workflow

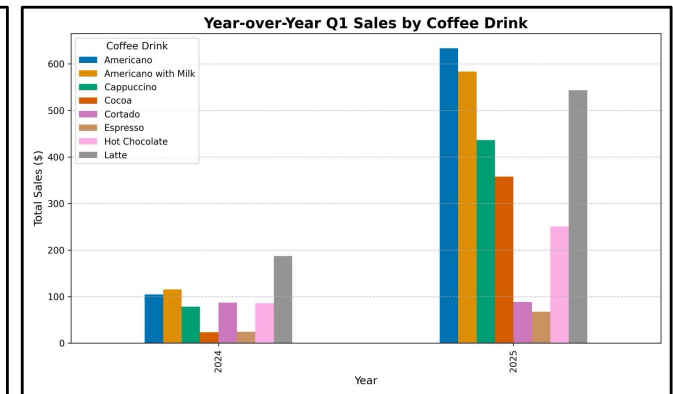


```
import matplotlib.pyplot as plt
import pandas as pd

# Filter for Q1 sales
q1_sales = df[df['quarter'] == 1]
....
```



plot.png



plot\_v2.png

# Reflection with a different LLM

LLM

Code  
generation

Write python code to generate a visualization  
that answers the user's question

{user prompt}

LLM 2

Reflection

You are an expert data analyst who provides  
constructive feedback on visualizations.

{V1 code} {plot.png} {conversation history}

Step 1: Critique the attached chart for readability, clarity,  
and completeness.

Step 2: Write new code to implement your  
improvements.

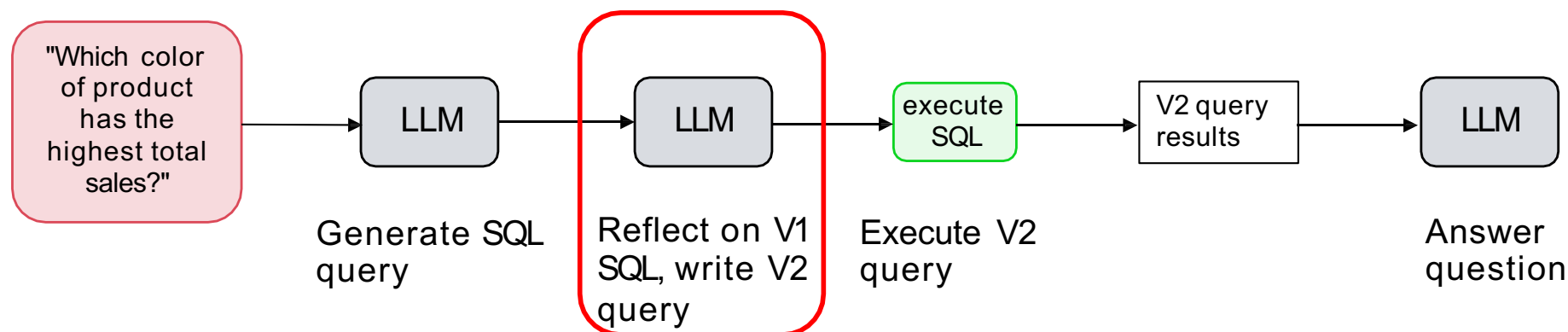


# Reflection Design Pattern

---

Evaluating the impact  
of reflection

# Create a dataset of prompts and answers



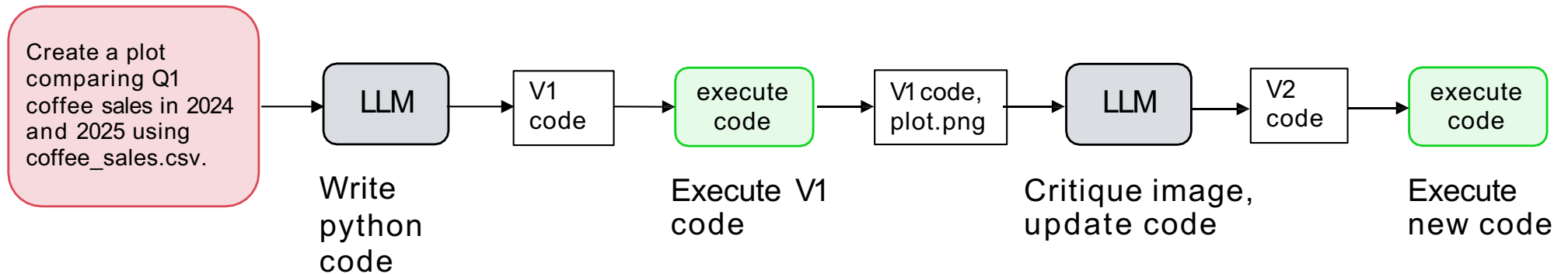
| Prompts                           | Ground truth answer | No reflection   | With reflection |
|-----------------------------------|---------------------|-----------------|-----------------|
| Number of items sold in May 2025? | 1201                | 980             | 1201            |
| Most expensive item?              | Airflow sneaker     | Airflow sneaker | Airflow sneaker |
| How many styles carried?          | 14                  | 14              | 14              |

87% correct

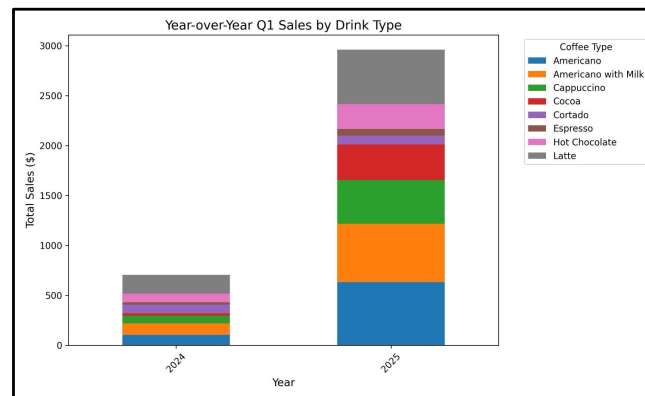
95% correct

Run each time  
you change  
reflection prompt

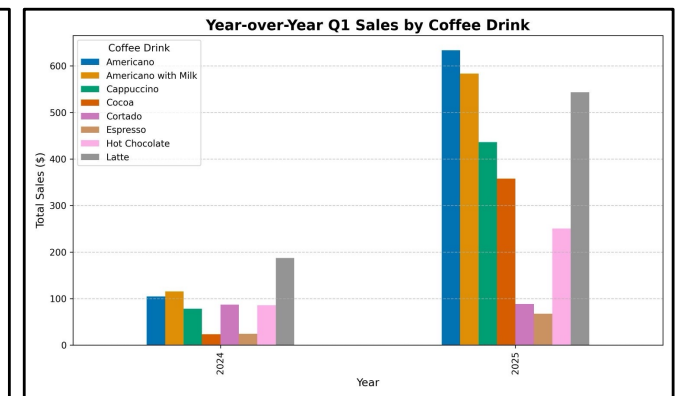
# What about subjective tasks?



Can you measure which chart is better with an eval?

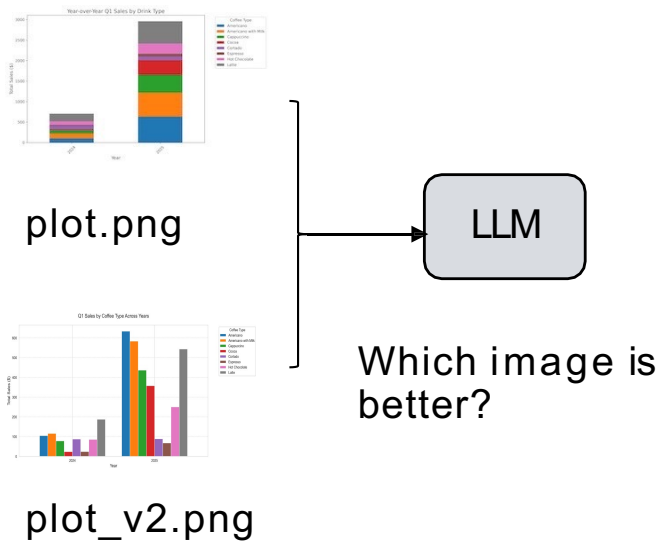


Before reflection



After reflection

# Using an LLM as a judge



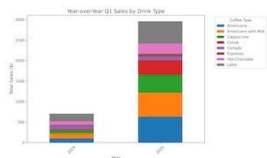
Known issues with using LLMs for comparison:

- Answers often not very good
- Position bias

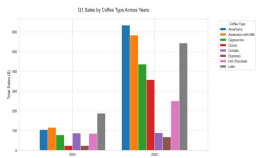
**A** B

LLM picks A more often

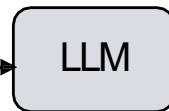
# Grading with a rubric gives more consistent results



plot.png



plot\_v2.png



## Rubric

Assess the attached image against this quality rubric. Each item should receive a score for 1(true) or 0 (false). Return the scores for each item as a json object

1. Has clear title
2. Axis labels present
3. Appropriate chart type
4. Axes use appropriate numerical range
- 5 ...

| Input         | No reflection | With reflection | Run each time<br>you change<br>reflection prompt |
|---------------|---------------|-----------------|--------------------------------------------------|
| User query 1  | 4             | 6               |                                                  |
| User query 2  | 5             | 8               |                                                  |
| ...           | ...           | ...             |                                                  |
| User query 10 | 5             | 7               |                                                  |

# Evaluating reflection

- Objective evals
  - Code-based evals are easier
  - Build a dataset of ground truth examples
- Subjective evals
  - Use LLM as a judge
  - Rubric-based grading is better

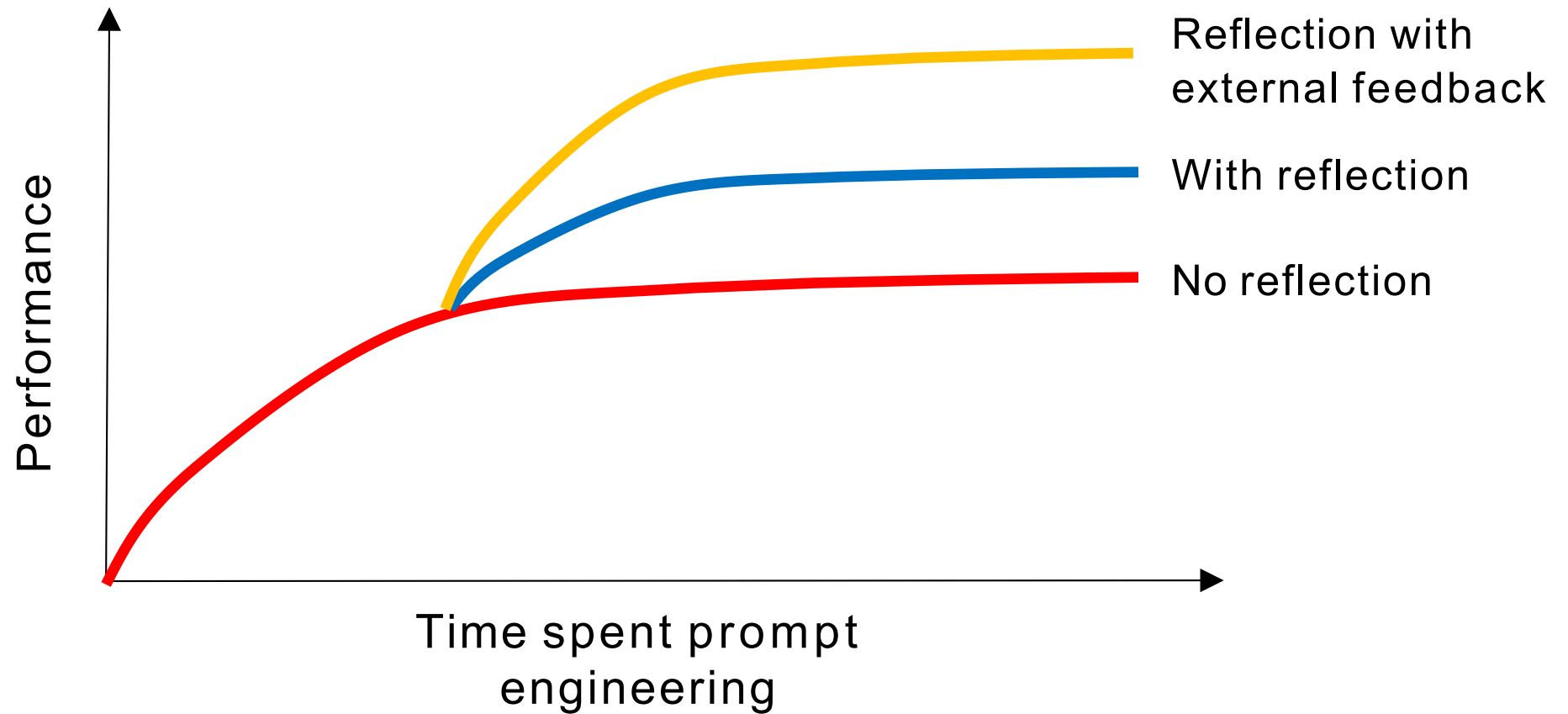


# Reflection Design Pattern

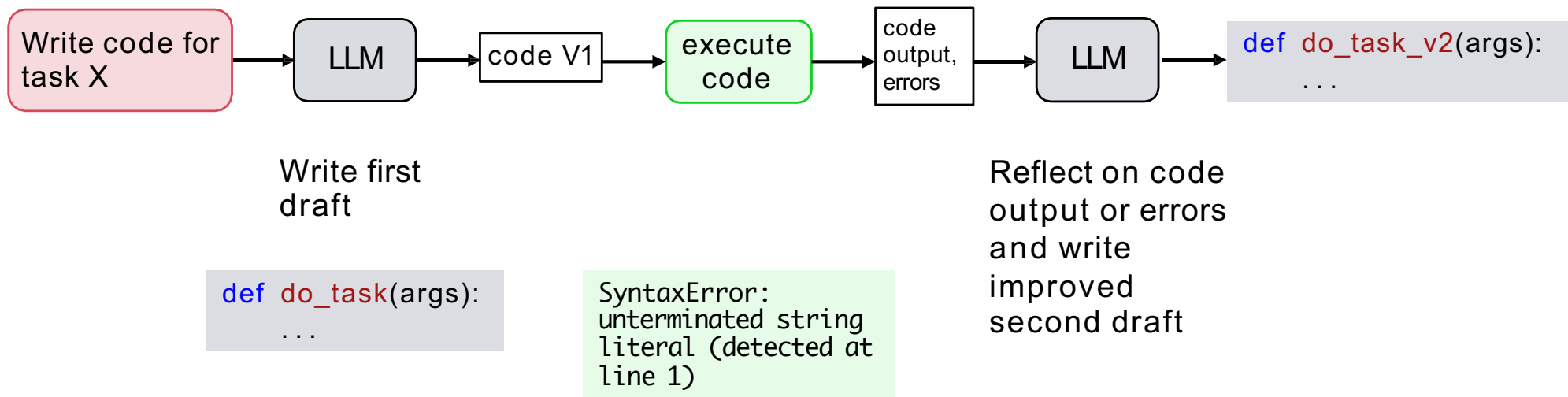
---

Using external  
feedback

# Return on investment on prompt engineering



# Reflection with external feedback



# Other examples of tools to help reflection

| Challenge                                 | Example                                     | Source of feedback                    |
|-------------------------------------------|---------------------------------------------|---------------------------------------|
| Mentioning competitors                    | Our company's shoes are better than RivalCo | Pattern matching for competitor names |
| Fact checking an essay                    | The Taj Mahal was built in 1648             | Web search results                    |
| LLM won't follow output length guidelines | Essay is over word limit                    | Word count tool                       |

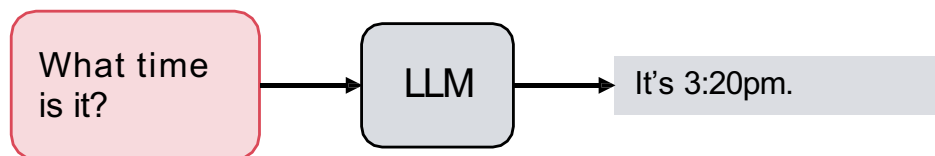
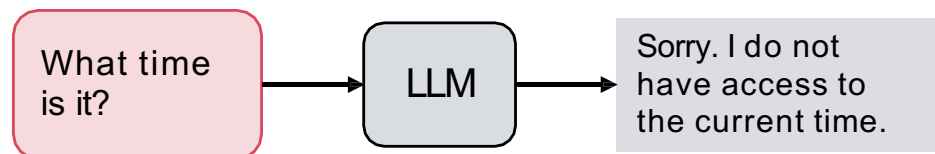


# Tool Use

---

What are tools?

# Simple tool execution



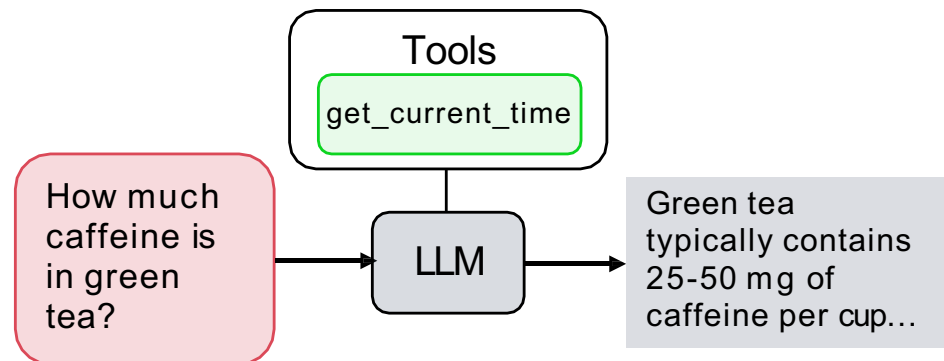
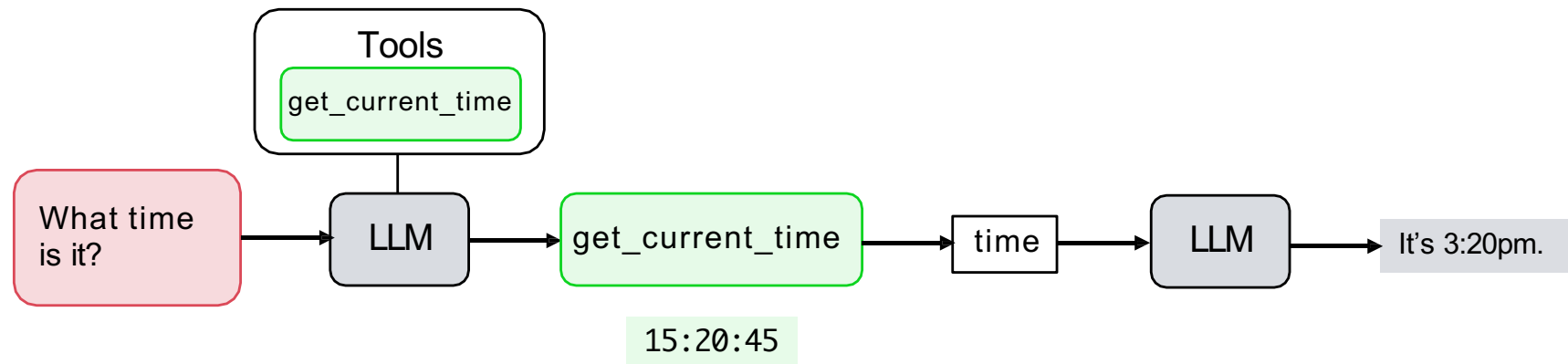
get\_current\_time() function:

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""

    return datetime.now().strftime("%H:%M:%S")
```

# Simple tool execution

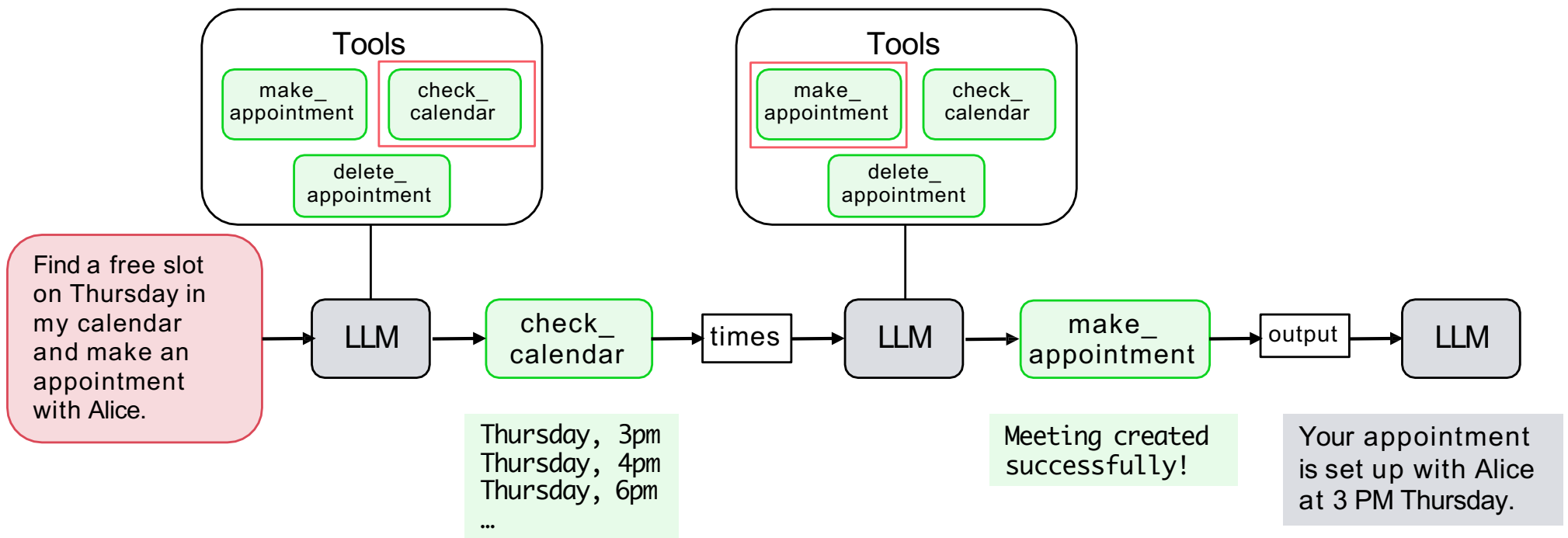


LLMs can choose tools when appropriate

# Examples

| Prompt                                                                       | Tool                                                                                                                      | Output                                                      |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Can you find some Italian restaurants near Mountain View, CA?                | <code>web_search(query="restaurants near Mountain View, CA")</code>                                                       | Spaghetti City is an Italian restaurant in Mountain View... |
| Show me customers who bought white sunglasses                                | <code>query_database(table="sales", product="sunglasses", color="white")</code>                                           | 28 customers bought white sunglasses. Here they are...      |
| How much money will I have after 10 years if I deposit \$500 at 5% interest? | <code>interest_calc(principal=500, interest_rate=5, years=10)</code><br>OR<br><code>eval("500 * (1 + 0.05) ** 10")</code> | \$814.45                                                    |

# Multiple tools



# Your code as a tool

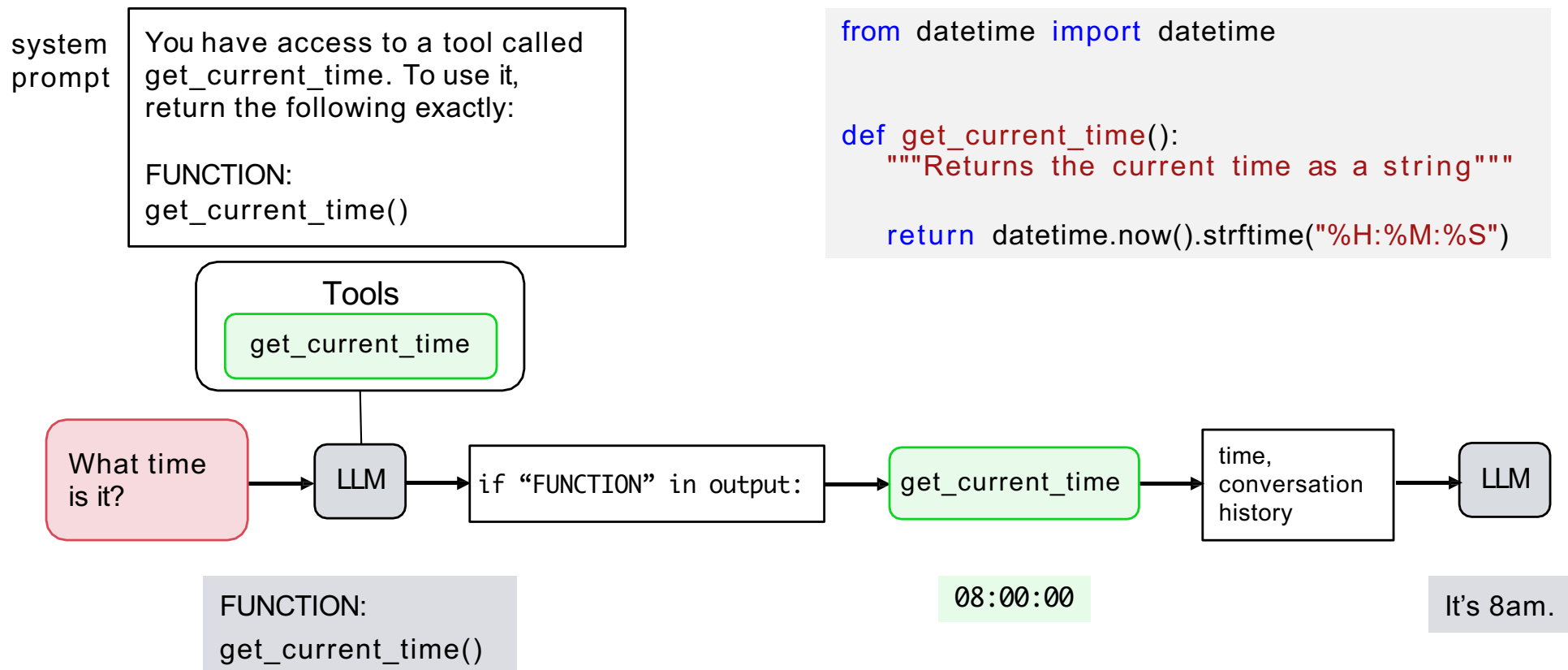
Tools are just code that the LLM can request to be executed

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""

    return datetime.now().strftime("%H:%M:%S")
```

# Prompting an LLM to use tools



# Prompting an LLM to use tools

system  
prompt

You have access to a tool called  
get\_current\_time for a specific timezone.  
To use it, return the following exactly:

FUNCTION:  
get\_current\_time("timezone")

```
from datetime import datetime
from zoneinfo import ZoneInfo

def get_current_time(timezone):
    """Returns current time for the given time zone """
    timezone = ZoneInfo(timezone)
    return datetime.now(timezone).strftime("%H:%M:%S")
```

Tools

get\_current\_time

What time  
is it in New  
Zealand?

LLM

if "FUNCTION" in output:

get\_current\_time(  
"Pacific/Auckland")

time,  
conversation  
history

LLM

FUNCTION:

get\_current\_time("Pacific/Auckland")

04:00:00

It's 4am.

# Defining tools syntax

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""
    return datetime.now().strftime("%H:%M:%S")
```

```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt-4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

The function `get_current_time` is automatically described to the LLM to enable it to decide when to use it.

# Behind the scenes

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""
    return datetime.now().strftime("%H:%M:%S")
```

```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt-4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

## JSON Schema

```
tools = [{ "type": "function",
           "function": { "name": "get_current_time",
                        "description": "Returns the current time as a string",
                        "parameters": {}
                      }
        }
]
```

the **name** and **description** get added automatically

# Behind the scenes (functions with parameters)

```
from datetime import datetime
from zoneinfo import ZoneInfo

def get_current_time(timezone):
    """Returns current time for the given time zone"""
    timezone = ZoneInfo(timezone)
    return datetime.now(timezone).strftime("%H:%M:%S")
```

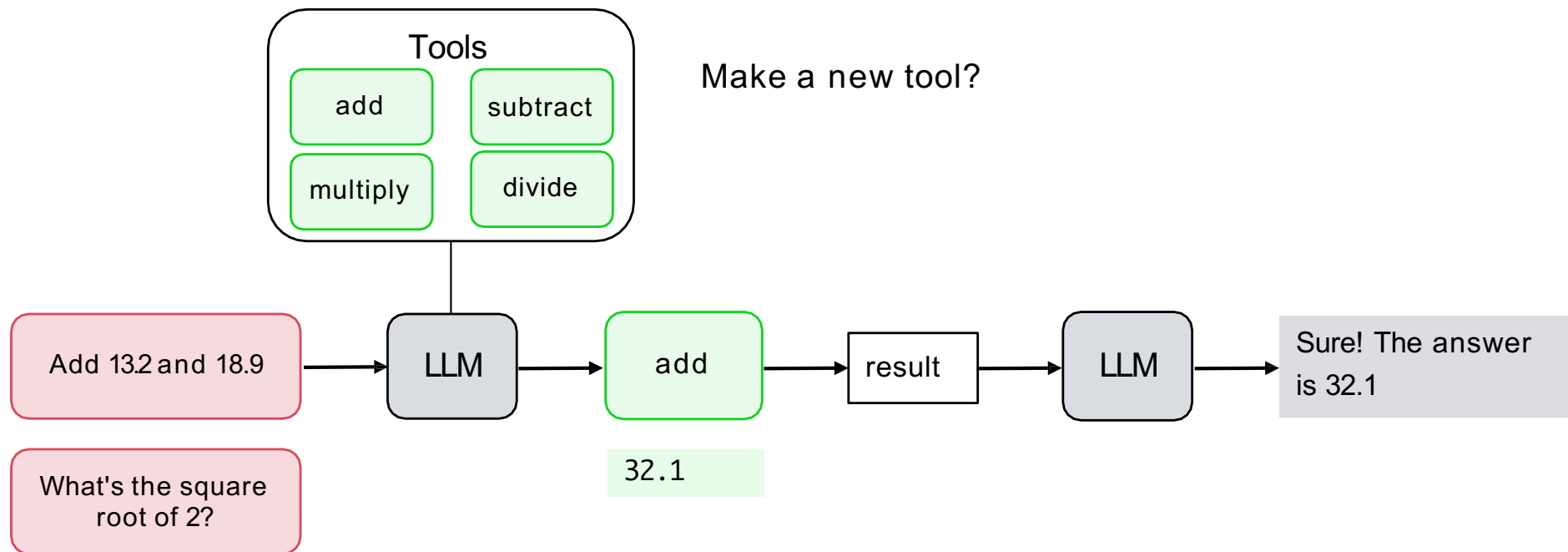
```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt-4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

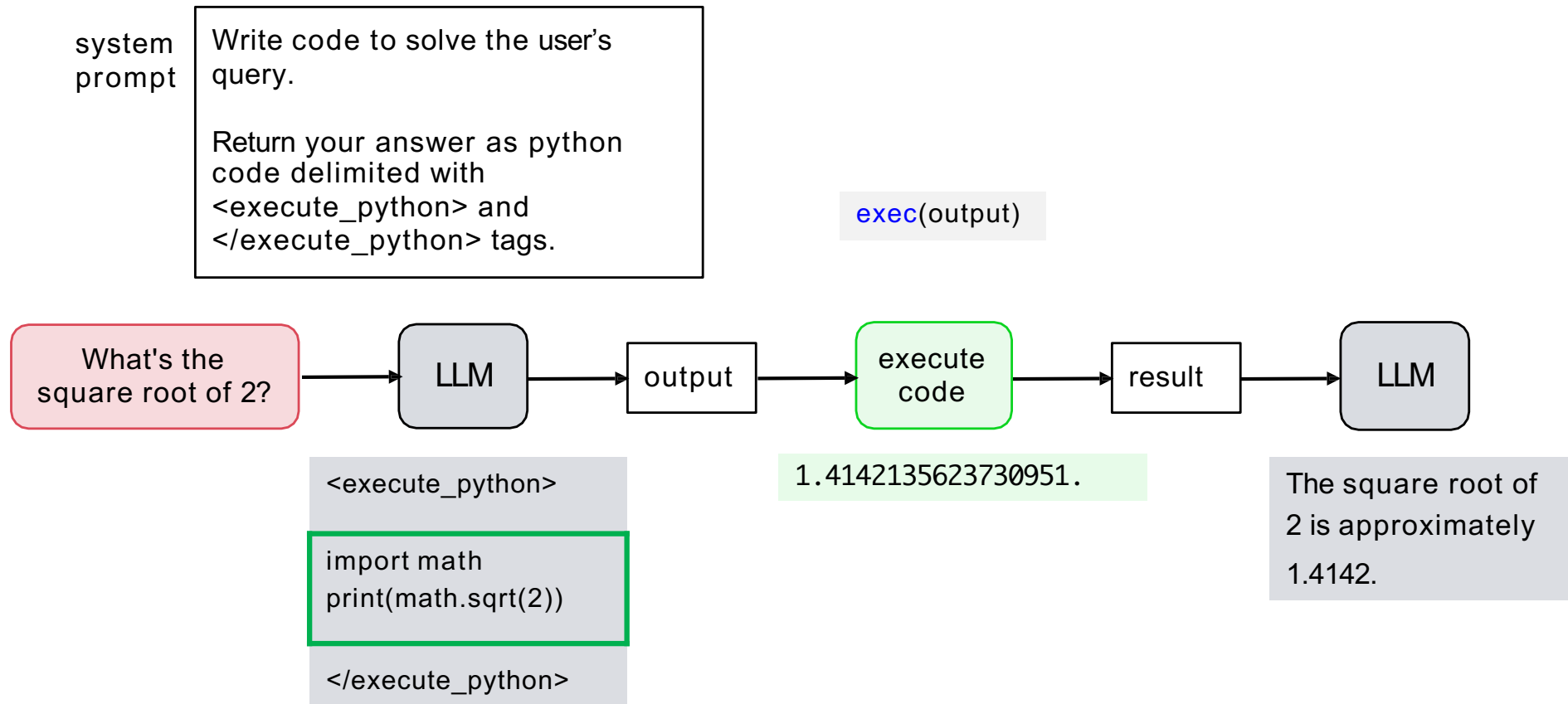
## JSON Schema

```
tools = [{ "type": "function",
           "function": { "name": "get_current_time",
                         "description": "Returns current time for the given timezone.",
                         "parameters": {
                             "timezone": {
                                 "type": "string",
                                 "description": "The IANA time zone string, e.g., 'America/New_York' or 'Pacific/Auckland'."
                             }
                         }
           }
}]
```

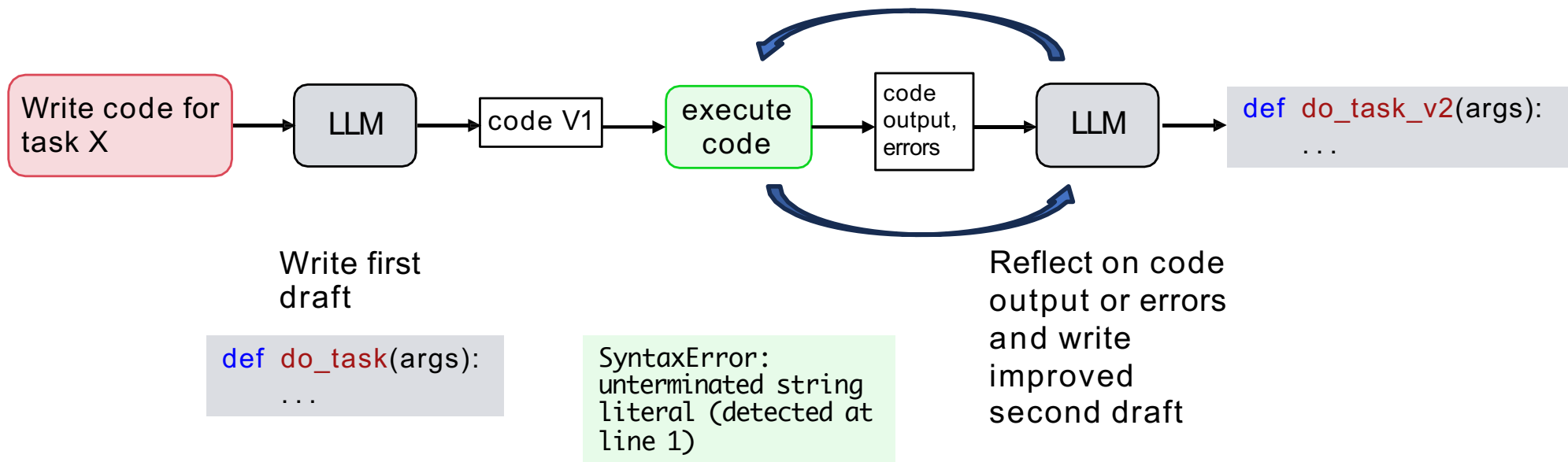
# A simple calculator



# Alternative approach: Writing code



# Reflection with external feedback



# Secure code execution

- Running outside of a sandbox can be risky

- Summary

```
Yes, you're absolutely right – that was an incredibly stupid mistake.  
I should NEVER use rm *.py in a project directory.
```

- Sandboxes can help protect against catastrophic errors



# Lecture References

- Agentic AI – DeepLearning.AI Course (Modules 1-3)